BlackSpot Miner: A Case Study-Based Data Mining Analysis of Severe Road Collisions

From Data to Decisions: Problem → Data → Preprocessing → Algorithm → Results → Evaluation → Insights → Decision

Group D15A_9	Domain	Case data	Reference paper	Date
Jaiprakash Talreja (54) Adarsh Singh (52) Vidushi Singh (53) Piyush Shukla (49)	Transportation and road safety	UK Department for Transport STATS19 collisions, 2021 to 2025	Kumar and Toshniwal (2015), Journal of Big Data 2:26	1 November 2026

1. Problem and business objective

1.1 Problem statement

Road crashes kill about 1.19 million people a year worldwide [1]. India alone recorded about 4.8 lakh accidents and 1.72 lakh deaths in 2023 [2]. UN Sustainable Development Goal target 3.6 calls for these deaths and injuries to be halved [3].

A road authority cannot rebuild every junction. It can fund a limited number of sites each year, so the practical problem is a choice: which places, and what should engineers examine when they get there? The usual approaches fall short in three ways.

- **Ranking by last year's crash count.** A bad year is partly bad luck, so sites picked this way tend to improve the next year even if nothing is done. The list then seems to work when it has not.
- **Treating all crashes alike.** A road with many minor shunts may need less attention than one with fewer crashes that kill or seriously injure people.
- **Reading one condition at a time.** Risk often sits in combinations, such as a motorcycle on a 60 mph road, which a single-column summary cannot show.

1.2 Business objective

Decision maker: the road safety team of a highway authority. **Objective:** spend a fixed safety budget where it covers the most severe crashes. Severe here means killed or seriously injured, written KSI throughout this report. The analysis has three goals.

Goal	Question	What the authority gets
Where	Which street-scale places concentrate KSI collisions?	A map of hotspots
Why	Which combinations of conditions raise the chance of KSI?	Tested rules that suggest what to review
What next	Which sites are likely to record KSI collisions next year?	A ranked priority list

Success test. The share of next year's KSI collisions that fall inside the top N ranked sites, compared with a list built from past crash counts alone.

1.3 Why this case and this data

The reference paper by Kumar and Toshniwal [4] mines 11,574 road accidents from Dehradun, India. It is open access and uses the syllabus methods of clustering and association rules, but it has no predictive model, no coordinates and no test on later data. India's eDAR (formerly iRAD) system now captures GPS-tagged accidents across all 36 States and UTs [5], but we could not find a public record-level download. Great Britain publishes every police-reported injury collision with coordinates [6], so we use that data to build and test the method. The pipeline is designed to run on eDAR records in the same way. The numbers themselves do not transfer to India.

2. Data understanding and source

Source: Department for Transport, Road Safety Open Data (STATS19), latest five years, published under the Open Government Licence v3.0 [6]. Police record each injury collision on a national form. Each row is a real, reported event.

513,801
collisions, 2021 to 2025

937,265
vehicle records

652,821
casualty records

8,033
people killed; 129,011 seriously injured

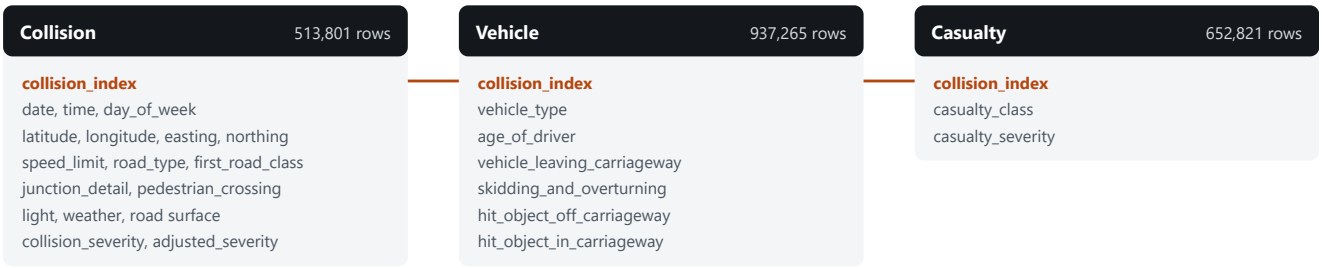


Figure 1. The three linked tables, joined on `collision_index`.

Attribute group	Attribute type	Use in the analysis
Location: easting, northing, latitude, longitude	Ratio (metres, degrees)	Spatial distance for hotspot discovery
Road: speed limit, class, layout, junction, crossing	Ordinal and nominal	Rule items and model features
Conditions: lighting, weather, surface, time, day	Nominal, plus derived time bands	Rule items and model features
People and vehicles: type, driver age, pedestrian hit	Nominal, binary and ratio	Rule items and model features
Target: collision severity	Ordinal (Fatal, Serious, Slight)	Grouped into KSI against Slight

A data problem that changes the design

Police forces are moving to injury-based reporting. The officer records the injury and the severity is derived from it. DfT's own early analysis found this switch added between 5 and 15% to serious injuries [7]. In this data, the injury-based share of collisions rose from 49.6% in 2021 to 86.5% in 2025, and 15 of 44 police force codes changed system during the period. Raw KSI share therefore partly measures a change of form. DfT publishes a probability for each collision that restates it as if every force used the injury-based system [7], and this analysis uses it.

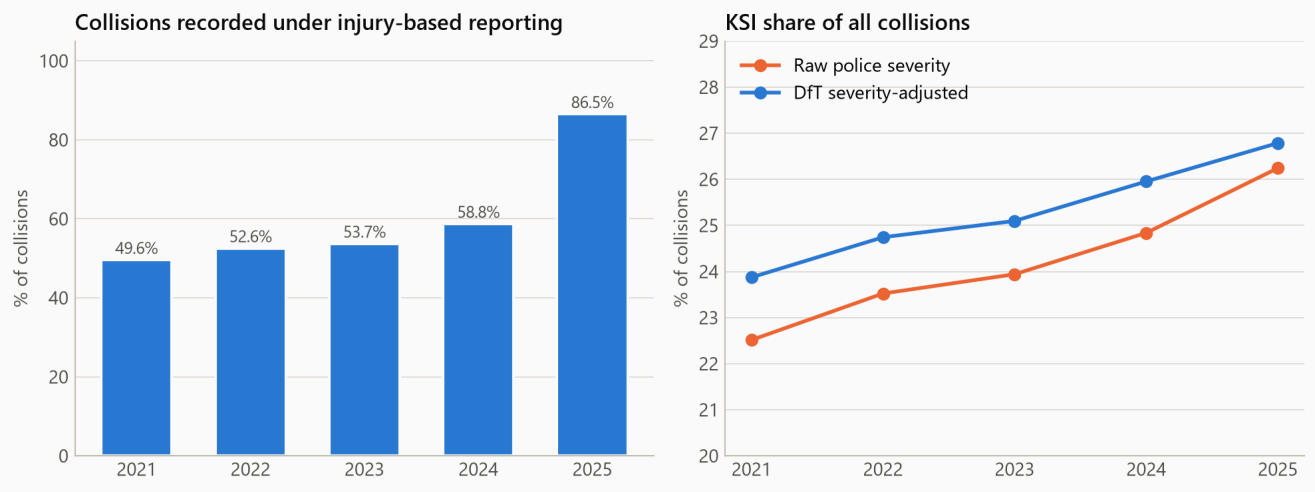


Figure 2. Reporting change. Raw KSI share rises from 22.51% to 26.24%. After adjustment the rise is smaller, 23.87% to 26.79%.

3. Data preprocessing (Module II)

Every field was audited year by year before use, because the 2024 STATS19 specification redefined several of them. Figure 3 shows the audit. Fields with an underlying coding change would teach a model the change of form, not road risk.

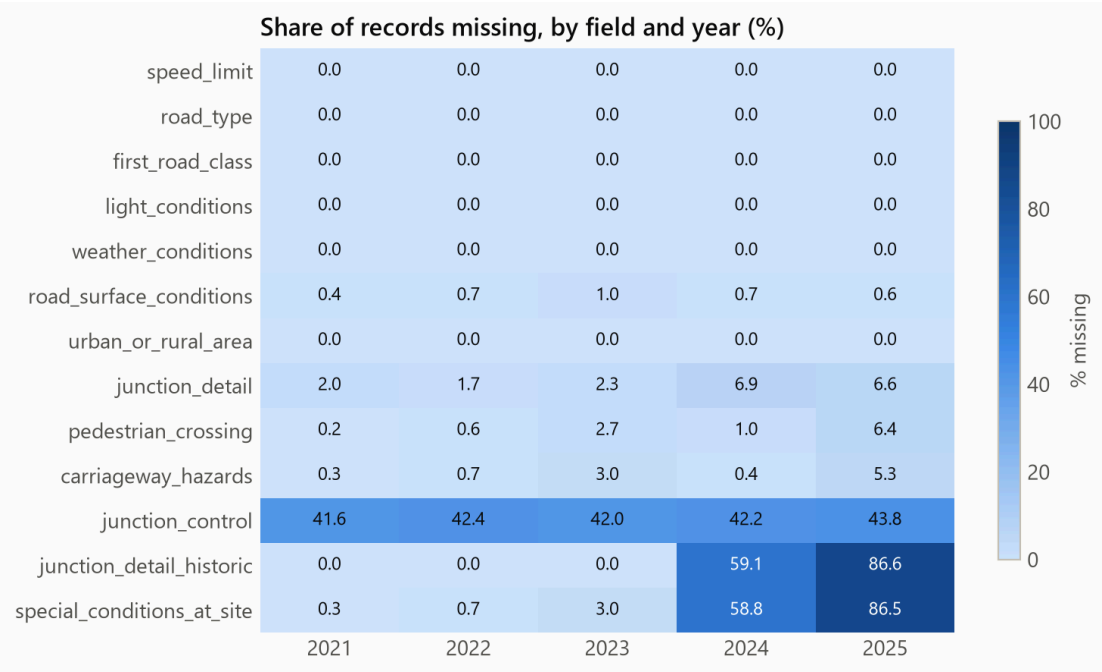


Figure 3. Field audit. `special_conditions_at_site` is 86.5% missing in 2025, so it is unusable even though earlier years look complete. High missing values for `junction_control` are structural: most collisions are not at a junction.

Step	What is done	Why
Data integration	Join vehicle and casualty tables to collisions; derive flags such as motorcycle involved, pedestrian hit, overturned	Who was involved is a stronger signal than road context
Data cleaning	Remove 56 records without a location or speed limit; map -1, 9 and 99 to an explicit Unknown	Unknown is not the same as a guessed value
Attribute subset selection	Drop `number_of_casualties`, police attendance and enhanced severity	They are known only after the outcome and would leak the label
Attribute subset selection	Drop `*_historic` fields and `special_conditions_at_site`	Mostly empty in recent years
Label consistency	Rules and severity models use only injury-based records (309,041). Location ranking uses DfT adjusted KSI on all records (513,745)	A serious injury must mean the same thing in every row
Data transformation	Map integer codes to labels from the official guide; one-hot encode for classifiers; item encode for Apriori	Codes are nominal and have no numeric order
Discretization	Bin hours into five bands; group weather, surface and crossing types	Rare categories cannot support rules
Aggregation	Build 500 m grid cells with three-year history features	Budgets are spent on places, not on single crashes
Split by time	Train on 2021 to 2024, test on 2025	A random split would let the model see the future

4. Identifying the data mining tasks

Business question	Data mining task	Type	Output	Module
Where do KSI collisions concentrate?	Density-based clustering	Descriptive	Hotspot clusters and noise	IV
Which combinations raise KSI risk?	Association rule mining	Descriptive	Rules with support, confidence, lift	V
Do crash types hide different patterns?	Categorical clustering, then rules per segment	Descriptive	Segments and their rules	IV, V
Is a collision likely to be KSI?	Binary classification	Predictive	Class and probability	III
Which sites will record KSI next year?	Classification and count prediction on grid cells	Predictive	Ranked sites	III
What should the authority do?	Decision support	Prescriptive use	Priority list	VI

5. Suitable algorithms and their justification

5.1 DBSCAN for hotspots [8]

How it works. A point is a core point if at least `min_samples` points lie within radius `eps`. Clusters grow by joining reachable core points, and anything unreachable stays as noise. **Why here.** Most severe collisions are isolated, and a black spot is a small dense patch of any shape. K-Means would force every collision into a cluster, needs k in advance and assumes round clusters of similar size. Earlier hotspot work used kernel density with K-Means [12]; DBSCAN gives each hotspot an explicit boundary without choosing k. **Limit.**

One `eps` cannot suit both central London and a rural road, and a large `eps` chains neighbouring junctions into one cluster, so settings are tested.

5.2 Apriori for association rules [9]

How it works. Each collision is a transaction of items such as *Speed: 60 mph*. Apriori uses the fact that every subset of a frequent itemset is frequent, which prunes the search. For a rule A → KSI:

`support = P(A and KSI)` `confidence = P(KSI | A)` `lift = P(KSI | A) / P(KSI)`

Why here. The attributes are categorical and the output is readable, so an engineer can act on it. It is also the method of the reference paper [4]. **Limit.** Rules show association, not cause.

5.3 K-Modes for segmentation [10]

How it works. Like K-Means, but it measures distance by the number of mismatched categories and represents each cluster by its most common values. **Why here.** Road accident data is mixed, and the reference paper reports that segmenting first reveals rules hidden in the whole data set [4]. Means have no meaning for categories such as lighting.

5.4 Decision Tree and Random Forest for classification [11]

Decision Tree. Splits greedily to reduce Gini impurity, `Gini(D) = 1 - sum(p_i^2)`. Trees have been used on accident severity before [13]. A pruned tree gives rules a traffic engineer can read and challenge. **Random Forest.** Averages many trees grown on bootstrap samples with random feature subsets, which reduces variance and captures interactions. It trades readability for accuracy, so both are used. Naive Bayes and AdaBoost are included as syllabus baselines.

Algorithm	Role	Strength here	Weakness here
DBSCAN	Where	Finds any shape, keeps noise out, no k needed	Sensitive to `eps`; one density for all regions
Apriori	Why	Readable, exhaustive at a set support	Many redundant rules; association only
K-Modes	Segments	Works directly on categories	Result depends on k and starting points
Decision Tree	Explain	White-box rules	Unstable; lower accuracy than ensembles
Random Forest	Predict	Handles interactions, resists overfitting	Less transparent; heavier to train

6. The data mining process



Figure 4. The eight-stage process applied to this case.

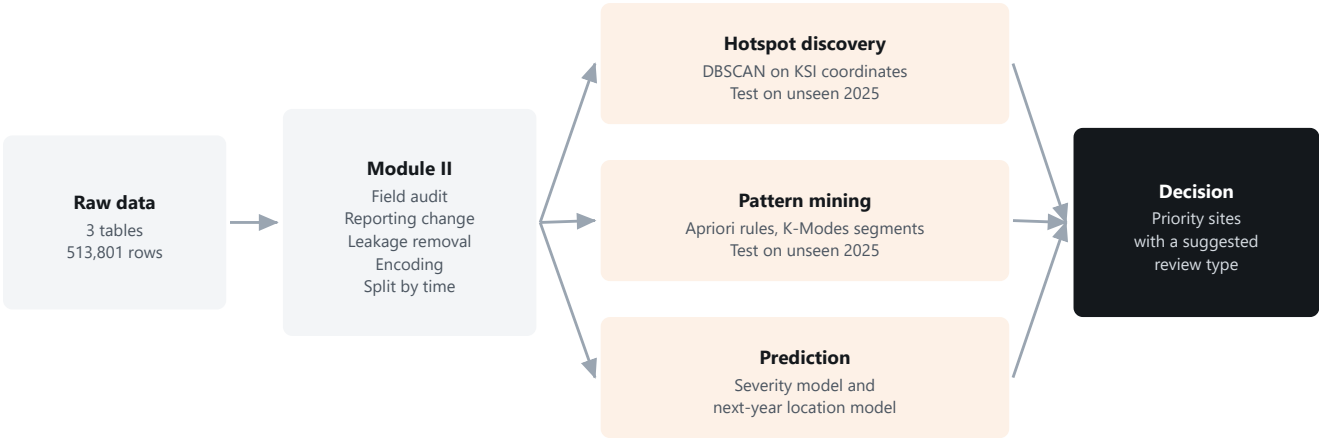


Figure 5. Technical architecture: one prepared data set feeds three branches, and all three feed the decision.

Stage	In this case study
Problem	Decide where a limited safety budget covers the most KSI collisions
Data	Five years of STATS19 collisions, vehicles and casualties
Preprocessing	Audit, integrate, handle the reporting change, remove leakage, encode, split by time
Algorithm	DBSCAN, Apriori, K-Modes, Naive Bayes, Decision Tree, Random Forest, AdaBoost
Results	Hotspots, rules, segments, severity probabilities, site scores
Evaluation	Test everything on 2025, a year never used for mining or training
Insights	Name the places and combinations that matter, and state what the data cannot show
Decision	Ranked site list, each with a suggested review type for an engineer

7. Expected and available results, and evaluation measures

7.1 Available results from the reference paper

On 11,574 Dehradun accidents from 2009 to 2014, Kumar and Toshniwal found six clusters [4]. They include two-wheeler accidents at intersections and curves (27.48%), two-wheeler accidents on highways near hills, forest and farmland (15.31%), falls from height (16.66%), multi-vehicle and fixed-object accidents with no light (12.04%), pedestrian hits (15.08%) and roll-overs (13.42%). Their main conclusion is that clustering before rule mining reveals information that rules from the entire data set do not.

7.2 Available results from our data exploration

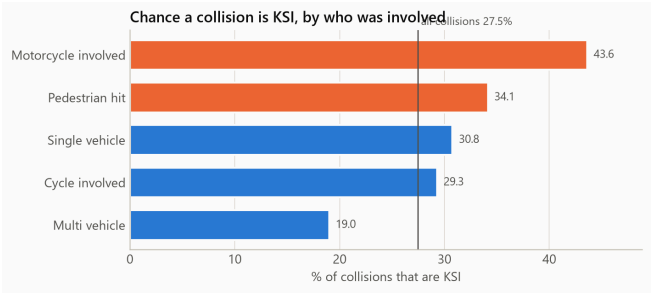


Figure 6. Motorcycle collisions are KSI 43.6% of the time, against 27.4% overall.

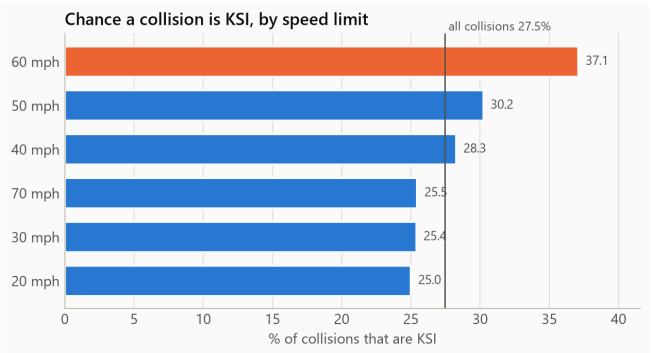


Figure 7. Severity peaks at 60 mph (37.1%), mostly rural single carriageways, not at 70 mph (25.5%), which is mostly motorway.

7.3 Evaluation measures

Measure	What it tells us	Why it fits this case
Recall, TP / (TP + FN)	Share of KSI collisions the model catches	Missing a severe crash pattern costs most
Precision, TP / (TP + FP)	Share of flagged collisions that really are KSI	Engineers' time is limited
F1	Balance of precision and recall	Accuracy alone misleads on a minority class
ROC-AUC	Chance a KSI case is ranked above a non-KSI case	Ranking is the real use
PR-AUC	Precision across recall levels; chance level equals the KSI rate	More honest than ROC-AUC when positives are rarer
Support, confidence, lift	How common, how reliable and how far above base rate a rule is	Same measures as the reference paper
Held-out rule lift	Whether a rule still holds on 2025	Rules that fail on new data are noise
Hotspot coverage	2025 KSI collisions reached, in total and per km^2	Shows whether hotspots persist
Budget curve	KSI collisions covered by the top N sites	This is the business success test

7.4 Expected results

- **Hotspots:** most KSI collisions should remain noise, with a small share in street-scale clusters, mostly in large cities.
- **Rules:** vulnerable road users and high-speed rural roads should lead, matching the two-wheeler patterns in the reference paper.
- **Severity model:** moderate performance, well below the figures common in tutorials, because severity depends on factors never recorded, such as impact speed and seat belt use.
- **Location model:** better than a past-count list, with a modest gain, because past counts are already a strong signal.

Measured results for all of these are in the CA2 report.

8. How data mining supports the final decision

Mining output	Decision it supports	Who acts
DBSCAN hotspots	Which places to inspect first	Road safety engineers
Rules by theme	What to check at a site: lighting, crossing, speed, junction layout	Engineers and police
Location model ranking	Next year's shortlist, and how many sites a budget covers	Budget holder
Severity model drivers	Which road users to target in campaigns	Road safety officers

What this analysis does not claim. It finds associations, not causes, so it cannot promise that a treatment will prevent a crash. STATS19 has no traffic volumes, so a busy site scores high partly because it is busy. Only police-reported injury collisions are included. The output is a shortlist for engineers to investigate, not an automatic decision.

Contribution to the goal. The authority moves from reacting to last year's counts to a tested, repeatable process that says where to look, what to look for and how much of next year's severe crash burden a given budget covers. This supports SDG target 3.6 on road deaths and injuries [3].

References

1. World Health Organization (2023). *Despite notable progress, road safety remains urgent global issue.* News release, 13 December 2023. https://www.who.int/news/item/13-12-2023-despite-notable-progress-road-safety-remains-urgent-global-issue
2. Press Trust of India (2025). *1.72 lakh people died in 4.80 lakh road accidents in 2023: Road Ministry report.* The Week, 28 August 2025. https://www.theweek.in/wire-updates/business/2025/08/28/del82-biz-road-accidents-report.html
3. United Nations Statistics Division. *SDG indicator metadata, target 3.6.* https://unstats.un.org/sdgs/metadata/files/Metadata-03-06-01.docx
4. Kumar, S. and Toshniwal, D. (2015). A data mining framework to analyze road accident data. *Journal of Big Data*, 2, 26. https://doi.org/10.1186/s40537-015-0035-y
5. Centre of Excellence for Road Safety, IIT Madras. *iRAD / e-DAR.* https://coers.iitm.ac.in/irad/
6. Department for Transport. *Road Safety Open Data.* Open Government Licence v3.0. https://www.gov.uk/government/statistical-data-sets/road-safety-open-data
7. Department for Transport. *Guide to severity adjustments for reported road casualties Great Britain.* https://www.gov.uk/government/publications/guide-to-severity-adjustments-for-reported-road-casualty-statistics/guide-to-severity-adjustments-for-reported-road-casualties-great-britain
8. Ester, M., Kriegel, H.-P., Sander, J. and Xu, X. (1996). A density-based algorithm for discovering clusters in large spatial databases with noise. *Proceedings of KDD-96*, 226-231.
9. Agrawal, R. and Srikant, R. (1994). Fast algorithms for mining association rules in large databases. *Proceedings of VLDB 1994*, 487-499.
10. Huang, Z. (1998). Extensions to the k-means algorithm for clustering large data sets with categorical values. *Data Mining and Knowledge Discovery*, 2(3), 283-304. https://doi.org/10.1023/A:1009769707641
11. Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5-32. https://doi.org/10.1023/A:1010933404324
12. Anderson, T. K. (2009). Kernel density estimation and K-means clustering to profile road accident hotspots. *Accident Analysis and Prevention*, 41(3), 359-364. https://doi.org/10.1016/j.aap.2008.12.014
13. Abellán, J., López, G. and de Oña, J. (2013). Analysis of traffic accident severity using Decision Rules via Decision Trees. *Expert Systems with Applications*, 40(15), 6047-6054. https://doi.org/10.1016/j.eswa.2013.05.027